

Israel Souza Ferreira

Data Scientist | Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Lattes](#)

PROFILE

Computer scientist with experience in **applied research and ML-enabled systems development**: experimentation, computer vision, APIs, and data. I support decisions with baselines, ablations, benchmarks, and automated testing.

PROFESSIONAL EXPERIENCE

Tieta.ai

R&D Researcher — Research and Innovation Training Program

Remote, Brazil

Jul 2026 - Present

- Developed **Python pipelines for deterministic segmentation** with lexical and syntactic analysis and OCR error recovery to structure educational questions.
- Implemented an API with **FastAPI** to enrich and audit Elasticsearch documents.
- Automated runs with **Docker**, artifact reuse, structural validation, and regression tests.
- Produced **documentation and audit artifacts** for traceability and technical governance; organized changes through version control to support reproducibility and controlled pipeline evolution.

Tieta.ai

Machine Learning Researcher — CNPq RHAЕ Research Fellow, part-time

Remote, Brazil

Apr 2025 - Jun 2026

- Developed and evaluated **multimodal, multitask models** to estimate psychometric characteristics of educational questions from textual and visual information, including **Item Response Theory** parameters.
- Ran ablations, paired comparisons, and statistical analyses to identify redundant components and configurations with better generalization; structured reproducible training and evaluation pipelines.
- Built LLM pipelines orchestrated by **DSPy** for structured analyses and investigated text-image semantic retrieval as an auxiliary representation.

Independent Consulting — Educational Institution

Full Stack Developer and Machine Learning Engineer

Remote, Brazil

2026 - Present

- Built the **Face Clock Evoluir**, a contracted facial-attendance system in final acceptance testing. Owned models, backend, database, security, testing, and preparation for deployment; the current delivery has **574 automated tests**.
- Implemented CPU-only 1:N recognition with YuNet + SFace, ONNX Runtime, and PostgreSQL/pgvector: **74 ms p95** with one identification in the benchmark. Open-set calibration reduced accepted unknowns from **45.4% to 0.2%** on public reference data.
- Experiments reduced **5 to 3 frames** without measured loss and showed **0.076 ms** exact search over 1,000 vectors; kept transactional pgvector search instead of adding FAISS without a demonstrated gain.
- Compared five approaches for medical-document triage; EfficientNet + Logistic Regression reached **87.5% accuracy** with seven fewer false positives at the same recall. For an 8 GB, GPU-less server, reduced images from **10.59 to 1.11 GB**, with no classifier decisions changed in the migration to ONNX Runtime.

Federal University of Ceará — Quixadá Campus

Teaching Assistant for Algorithms, Calculus, and Pre-Calculus

On-site, Brazil

2022 - 2024

- Supported classes in data structures, algorithms, complexity, and Calculus, translating quantitative material across skill levels.

SELECTED PROJECTS

- Triple Roman Domination in Graphs (undergraduate thesis)**: implemented GA- and ACO-based metaheuristics and proposed a correction to the published ILP formulation; evaluated over **362 graphs**.
- Brazilian Emergency Aid — Data Engineering**: chunked binary COPY ingestion over **31.6 GB** of public data; instrumentation linked memory growth to retained deduplication IDs (**r = 0.99**).

TECHNICAL SKILLS

Languages and querying: Python, SQL, C++, and Bash.

ML, vision, and experimentation: PyTorch, Scikit-learn, OpenCV, YuNet, SFace, ONNX Runtime, DSPy, Optuna, cross-validation, calibration, ablation, hypothesis testing, and vector search.

Data and application engineering: PostgreSQL, pgvector, FAISS, Pandas, NumPy, SciPy, FastAPI, SQLAlchemy, Alembic, Docker, GitHub Actions, pytest, and Streamlit.

EDUCATION

Federal University of Ceará — Quixadá Campus

B.Sc. in Computer Science

Brazil

Completed Aug 19, 2026

LANGUAGES

Portuguese: Native | **English**: Fluent technical reading of research papers and documentation